Gianmarco Cafferata

Padron: 99423
1er cuatrimestre 2020

1)

$$y[n+2] + y[n+1] - \frac{3}{4}y[n] = x[n+1] + x[n]$$

a) Aplico transformada Z a ambos lados

$$Y(z) \cdot \left[z^2 + z - \frac{3}{4} \right] = X(z)\left[z + 1 \right]$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{z+1}{z^2 + z - \frac{3}{4}}$$

Numerador: $z + 1$

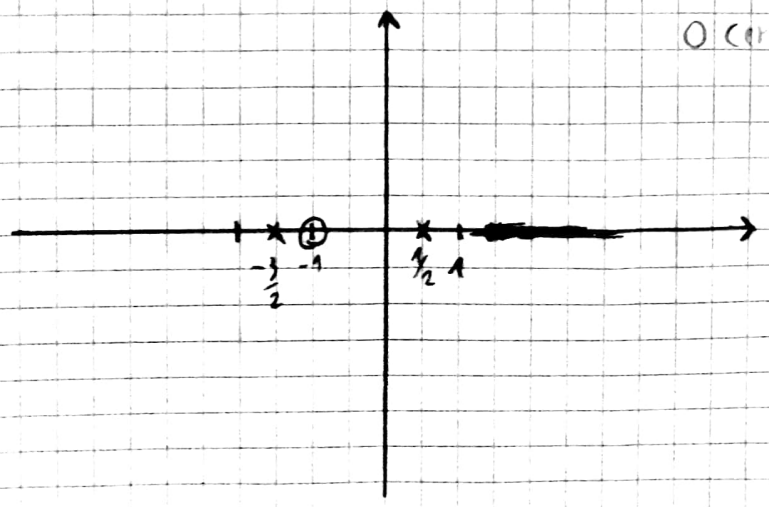
Denominador: $z^2 + z - \frac{3}{4} = \left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)$

$$H(z) = \frac{z+1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 + \frac{3}{2}z^{-1}\right)}$$

Ceros: -1

Polos: $\frac{1}{2}$ y $-\frac{3}{2}$

Hay 3 ROCs posibles, diagrama de Polos y ceros:



Asamblea
Escaneado con CamS

ROC 1 = $\{z \in \mathbb{C} / |z| < \frac{1}{2}\}$ No es causal ni estable

ROC 2 = $\{z \in \mathbb{C} / \frac{1}{2} < |z| < \frac{3}{2}\}$ Estable no causal

ROC 3 = $\{z \in \mathbb{C} / |z| > \frac{3}{2}\}$ Causal no estable

b) Las exponenciales son autofunciones, por propiedad

$$y[n] = H(2) \cdot 2^n = \frac{2+1}{\left(1-\frac{1}{2}\cdot\frac{1}{2}\right)\left(1+\frac{3}{2}\cdot\frac{1}{2}\right)} 2^n = \frac{16}{7} 2^n$$

Valido para todas las ROCs

Escaneado con CamS

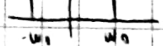
2) $r_c(t) = \cos^2(\omega_0 t)\cos^2(2\omega_0 t)$

Por propiedad del producto:

$$R_c(j\omega) = \mathcal{F}\left[\cos^2(\omega_0 t)\cos^2(2\omega_0 t)\right] = \mathcal{F}\left[\cos^2(\omega_0 t)\right] * \mathcal{F}\left[\cos^2(2\omega_0 t)\right]$$

$$\mathcal{F}\left[\cos^2(\omega_0 t)\right] = \mathcal{F}\left[\cos(\omega_0 t)\right] * \mathcal{F}\left[\cos(\omega_0 t)\right]$$

$$= \left[\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]\right] * \left[\pi\left[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)\right]\right]$$





$$= \pi^2\left[\delta(\omega-\omega_0) * \delta(\omega-\omega_0) + 2\,\delta(\omega-\omega_0) * \delta(\omega+\omega_0) + \delta(\omega+\omega_0) * \delta(\omega+\omega_0)\right]$$

Linealidad de convolución

$$= \pi^2\left[\delta(\omega-2\omega_0) + 2\,\delta(\omega) + \delta(\omega+2\omega_0)\right] = \mathcal{F}\left[\cos^2(\omega_0 t)\right]$$

Usando eso

$$\mathcal{F}\left[\cos^2(2\omega_0 t)\right] = \pi^2\left[\delta(\omega-4\omega_0) + 2\,\delta(\omega) + \delta(\omega+4\omega_0)\right]$$

$$R_c(j\omega) = \mathcal{F}\left[\cos^2(\omega_0 t)\right] * \mathcal{F}\left[\cos^2(2\omega_0 t)\right]$$

$$= \pi^2\Big[\delta(\omega-2\omega_0) \overset{①}{*} \mathcal{F}\left[\cos^2(2\omega_0 t)\right] + 2\,\delta(\omega) \overset{②}{*} \mathcal{F}\left[\cos^2(2\omega_0 t)\right] + \delta(\omega+2\omega_0) \overset{③}{*} \mathcal{F}\left[\cos^2(2\omega_0 t)\right]\Big]$$

Asamblea

Escaneado con CamS

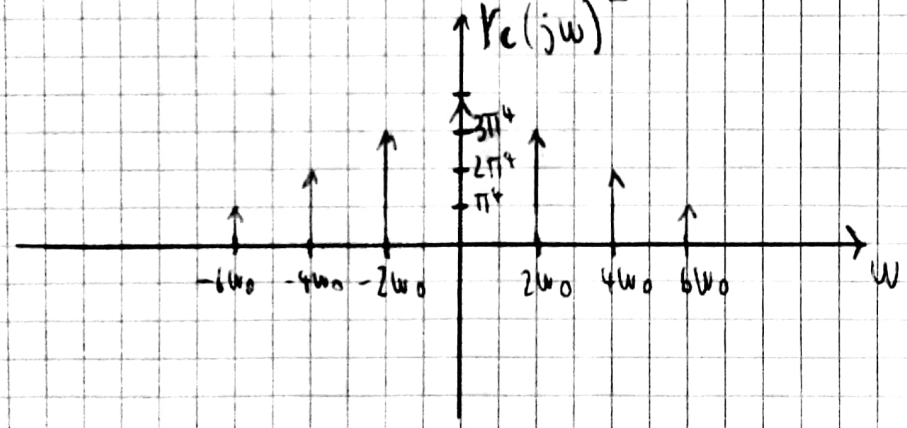
$$① = \pi^2 \left[\delta(w - 6w_0) + 2\delta(w - 2w_0) + \delta(w + 2w_0) \right]$$

$$② = \pi^2 \left[2\delta(w - 4w_0) + 4\delta(w) + 2\delta(w + 4w_0) \right]$$

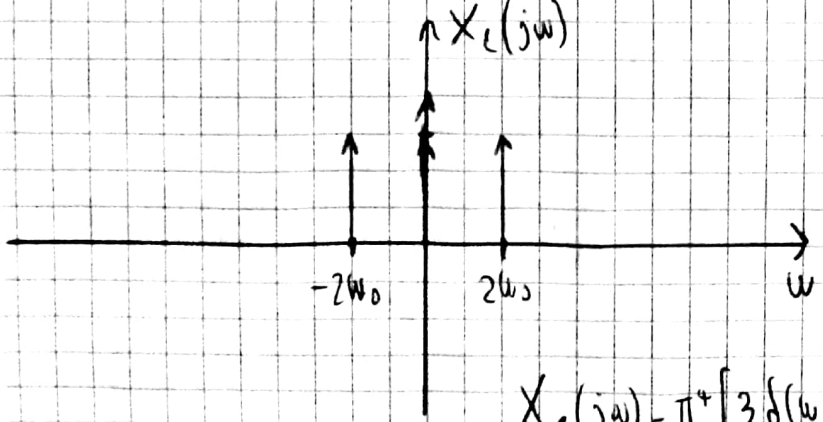
$$③ = \pi^2 \left[\delta(w - 2w_0) + 2\delta(w + 2w_0) + \delta(w + 6w_0) \right]$$

$$Y_c(jw) = \pi^2 \left[① + ② + ③ \right]$$

$$= \pi^4 \left[\delta(w - 6w_0) + 2\delta(w - 4w_0) + 3\delta(w - 2w_0) + 4\delta(w) + 3\delta(w + 2w_0) + 2\delta(w + 4w_0) + \delta(w + 6w_0) \right]$$



$x_c(t)$ es $y_c(t)$ Pasa por un pasabajos con corte $3w_0$.

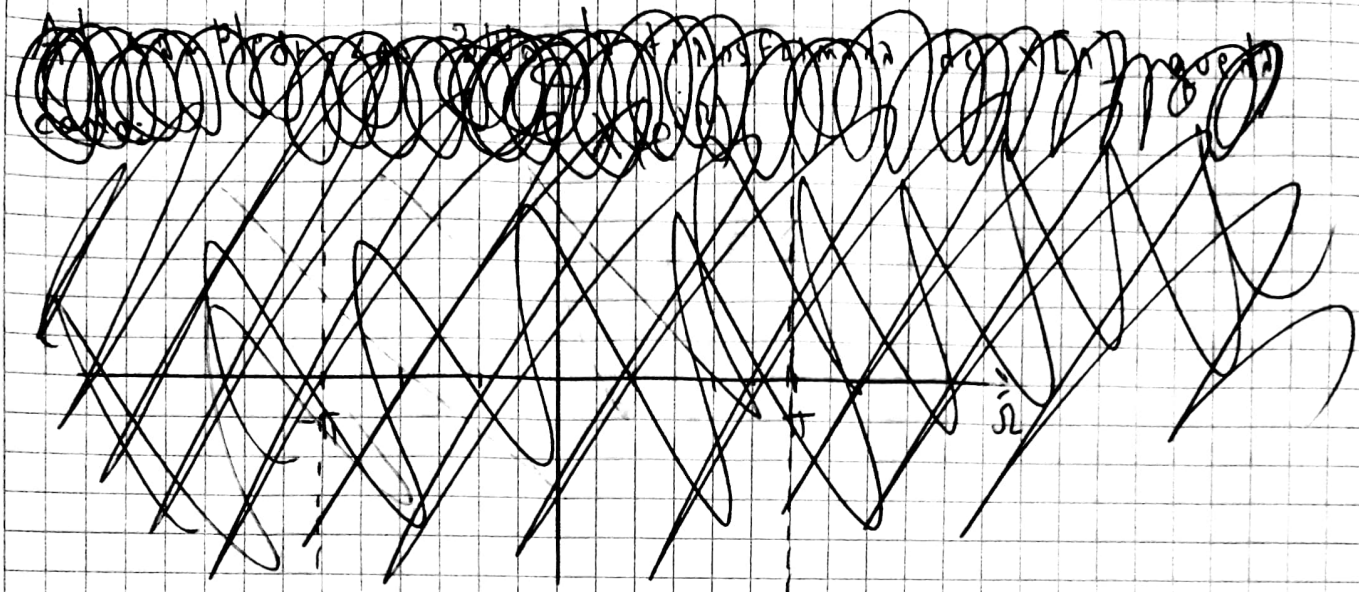


$$X_c(jw) = \pi^4 \left[3\delta(w - 2w_0) + 4\delta(w) + 3\delta(w + 2w_0) \right]$$

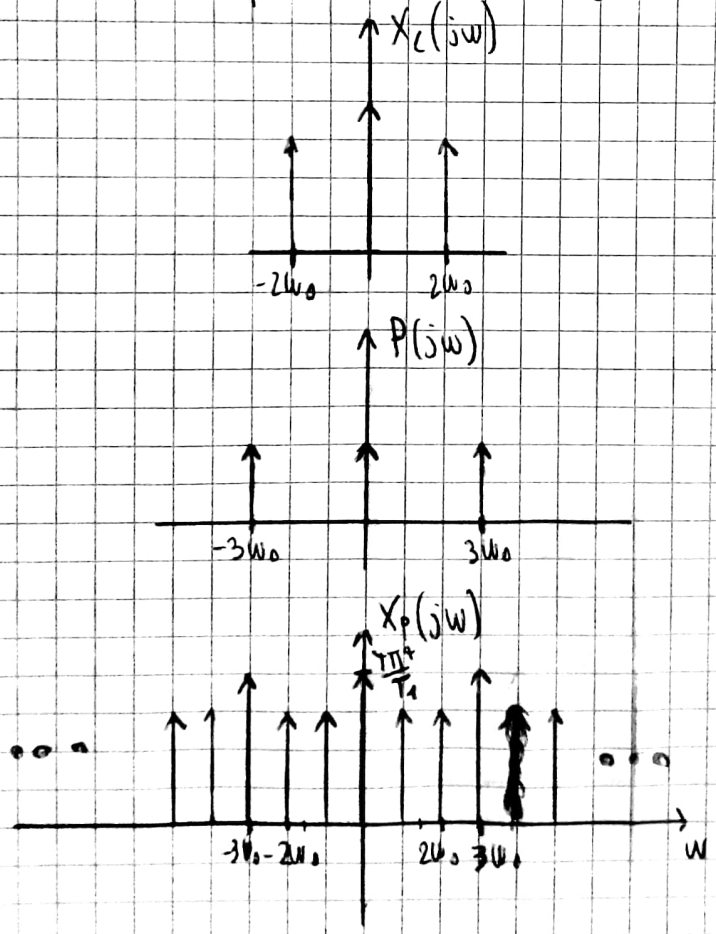
Asamblea

Escaneado con CamS

$T_1 = \frac{2\pi}{3\omega_0} \Rightarrow F_1 = \frac{3\omega_0}{2\pi} \Rightarrow \omega_1 = 3\omega_0$



Al no cumplir Nyquist hay aliasing: $(\omega_s = 3\omega_0)$

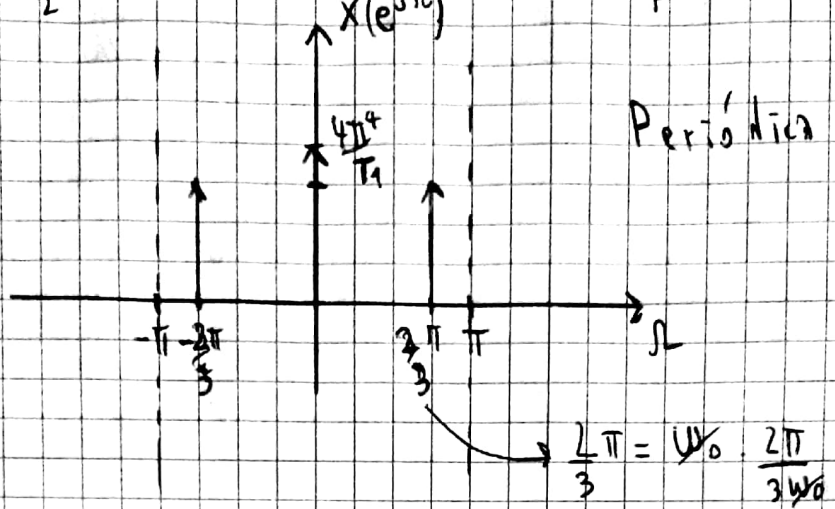


Asamblea

Escaneado con CamS

$\Omega = \pm\pi$ se corresponde con $\pm \frac{\omega_s}{2} = \frac{3}{2}\omega_0$

Nos quedamos para $x[n]$ con el espectro de X_p entre $-\frac{3}{2}\omega_0$ y $\frac{3}{2}\omega_0$ y para $X(e^{j\Omega})$ queda entre $-\pi$ y π escalado $\frac{1}{T_1}$

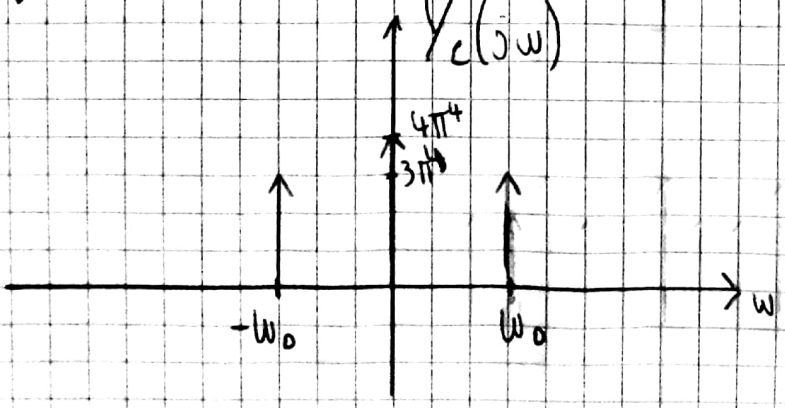


Como $H(e^{j\Omega})$ no filtra nada y es de ganancia 1:

$$Y(e^{j\Omega}) = X(e^{j\Omega})$$

$Y[n]$ pasa por D/A ideal con $T_2 = T_1$, filtra $Y(e^{j\Omega})$ entre $-\pi$ y π y escala en tiempo dividiendo por T_2

Con ganancia T_2



$\frac{2\pi}{3} \cdot \frac{1}{T_2} = \frac{2\pi}{3} \frac{3\omega_0}{2\pi} = \omega_0$

Asamblea

Escaneado con CamS

$$Y_c(j\omega) = \pi^2\left[3\delta(\omega-\omega_0) + 4\delta(\omega) + 3\delta(\omega+\omega_0)\right]$$

$$= \pi^2\left[\underbrace{\pi^2\left[\delta(\omega-\omega_0) + 2\delta(\omega) + \delta(\omega+\omega_0)\right]}_{\mathcal{F}\left[\cos^2\left(\frac{\omega_0}{2}t\right)\right]} + \underbrace{\frac{\pi^2}{2}\left[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)\right]}_{\frac{\pi}{2}\,\mathcal{F}\left[\cos(\omega_0 t)\right]}\right]$$

$$Y_c(f) = \pi^2\left[\cos^2\left(\frac{\omega_0}{2}t\right) + \frac{\pi}{2}\cos(\omega_0 t)\right]$$

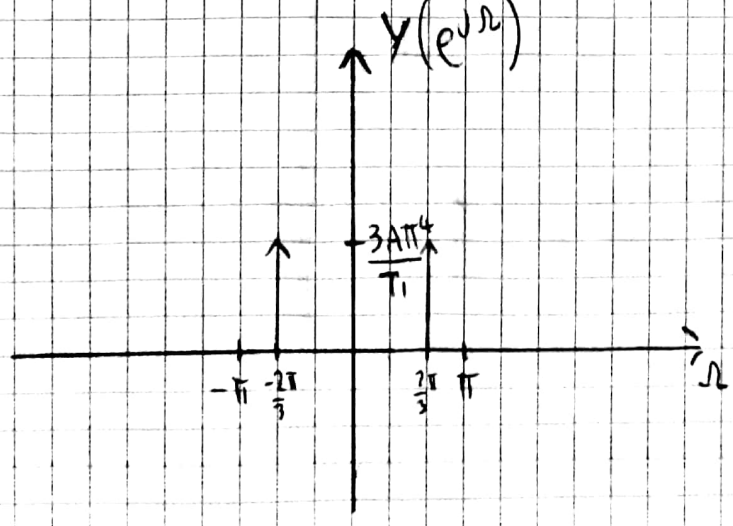
b) $y_c(t) = \cos(2\omega_0 t)$

$$Y_c(j\omega) = \pi\left[\delta(\omega-2\omega_0) + \delta(\omega+2\omega_0)\right]$$

Propongo:

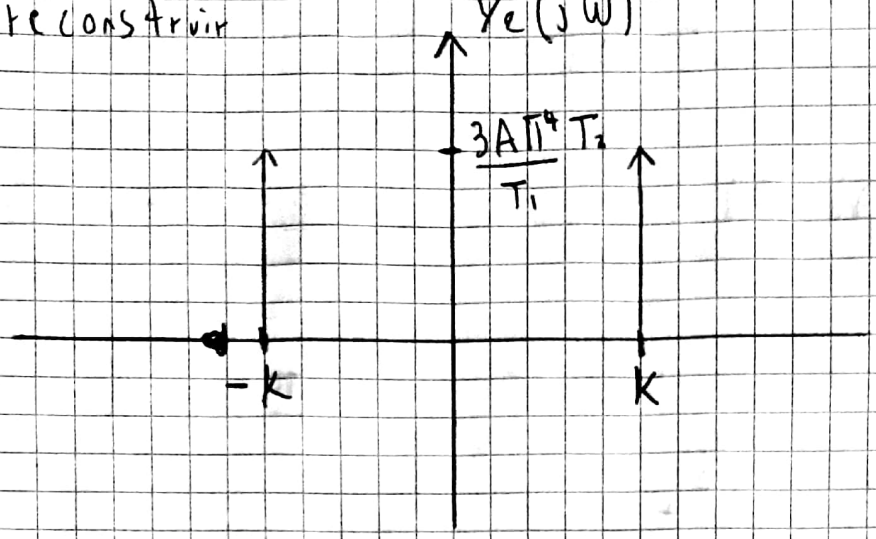
$$H(e^{j\Omega}) = \begin{cases} A & \text{si } |\Omega| \in \left[\frac{1}{3}\pi, \pi\right], \quad A \in \mathbb{R} \\ 0 & \text{e. o. c} \end{cases}$$

Entonces:



Asamblea
Escaneado con CamS

Al reconstruir



$$K = \frac{2\pi}{3}\frac{1}{T_2}$$

Quiero $K = 2w_0 = \frac{2\pi}{3}\frac{1}{T_2} \Rightarrow \boxed{T_2 = \frac{2\pi}{3}\frac{1}{2w_0} = \frac{2\pi}{6w_0}}$

Y $\frac{3A\pi^4}{T_1} T_2 \cdot = \pi \Rightarrow \frac{3A\pi^4}{\frac{2\pi}{3w_0}} \cdot \frac{2\pi}{6w_0} = \pi$

$3A\pi^4 \cdot \frac{\cancel{2\pi}}{6\cancel{w_0}} \cdot \frac{3\cancel{w_0}}{\cancel{2\pi}} = \frac{3}{6} \cdot 3\pi^4 \cdot A = \pi \Rightarrow A = \frac{\pi}{\pi^4}\cdot\frac{6}{9} = \frac{6}{9\pi^3}$

$$\boxed{H(e^{j\Omega}) = \begin{cases} \frac{6}{9\pi^3} & \text{si } |\Omega| \in \left[\frac{1}{3}\pi, \pi\right] \\ 0 & \text{e.o.c} \end{cases}}$$

Escaneado con CamS

3)

a)

$$X_1[k] = \sum_{n=0}^{N-1} x_1[n] W_N^{kn} = \sum_{n=0}^{N-1} x_1[n] e^{-j\frac{2\pi}{N}kn}$$

$$X_2[k'] = \sum_{n=0}^{N-1} x_2[n] W_N^{kn} = \sum_{n=0}^{N-1} x_2[n] e^{-j\frac{2\pi}{N}k'n}$$

Por propiedad

$$x^*[(-n)]_N \xleftrightarrow{DFT} X^*[k]$$

$$X_2^*[k'] = \sum_{n=0}^{N-1} x_2^*[n] e^{j\frac{2\pi}{N}k'n}$$

$$X_1[k] X_2^*[k'] = \sum_{n=0}^{N-1} x_1[n] \sum_{n'=0}^{N-1} x_2^*[(-n')_N] e^{j\frac{2\pi}{N}(k-k')n}$$

$$\sum_{k=0}^{N-1} X_1[k] X_2^*[k] = \sum_{n=0}^{N-1} x_1[n] \sum_{n'=0}^{N-1} x_2^*[(-n')_N] \sum_{k=0}^{N-1} e^{j\frac{2\pi}{N}k(n-n')}$$

$$X_1[k] X_2^*[k] = \sum_{n=0}^{N-1} x_1[n] \sum_{n'=0}^{N-1} x_2^*[(-n')_N] e^{-j\frac{2\pi}{N}k(n-n')}$$

$$\sum_{k=0}^{N-1} X_1[k] X_2^*[k] = \sum_{n=0}^{N-1} x_1[n] \sum_{n'=0}^{N-1} x_2^*[(-n')_N] \sum_{k=0}^{N-1} e^{-j\frac{2\pi}{N}k(n-n')}$$



Escaneado con CamS

$$x_1[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_1[k] W_N^{-kn}$$

$$x_2^*[n] = \frac{1}{N} \sum_{k'=0}^{N-1} X_2^*[((-k'))_N] W_N^{+k'n}$$

$$x_1[n]\, x_2^*[n] = \frac{1}{N^2} \sum_{k=0}^{N-1} X_1[k] \sum_{k'=0}^{N-1} X_2^*[((-k'))_N] W_N^{-n(k-k')}$$

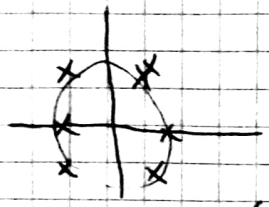
$$\sum_{n=0}^{N-1} x_1[n]\, x_2^*[n] = \frac{1}{N^2} \sum_{k=0}^{N-1} X_1[k] \sum_{k'=0}^{N-1} X_2^*[((-k'))_N] \underbrace{\sum_{n=0}^{N-1} W_N^{-n(k-k')}}_{?\ *}$$

$*$

$$\sum_{n=0}^{N-1} W_N^{-n(k-k')} = \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-k')n}$$

Si $k = k'$ $\quad \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(0)\cdot n} = N$

Si $k \neq k'$



Suma puntos sobre $|z| = 1$ que se anulan

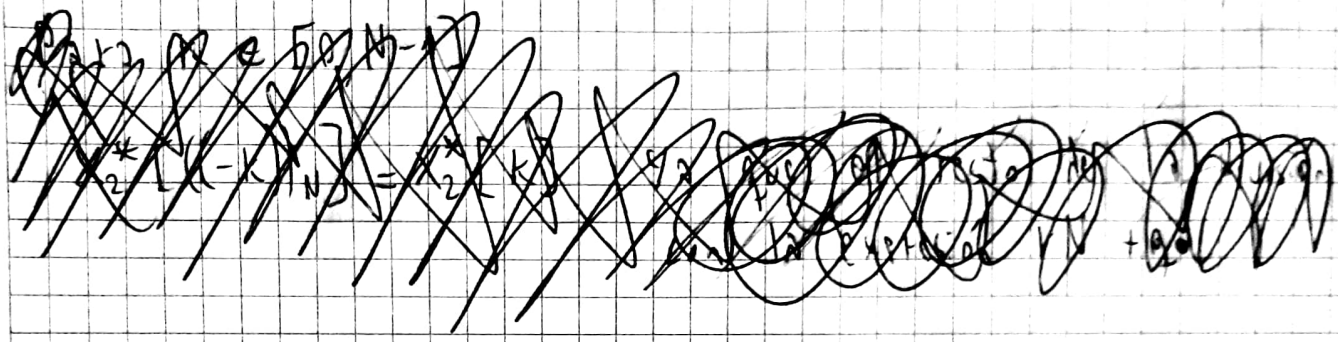
$$\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-k')n} = \begin{cases} N & \text{si } k = k' \\ 0 & \text{e. o. l} \end{cases}$$

Asamblea

Escaneado con CamS

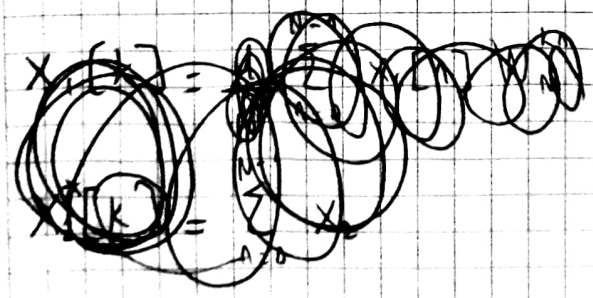
$$\sum_{n=0}^{N-1} x_1[n]\, x_2^*[n] = \frac{N}{N^2} \sum_{k=0}^{N-1} x_1[k]\, x_2^*[((-k))_N]$$

$$\sum_{n=0}^{N-1} x_1[n]\, x_2^*[n] = \frac{1}{N} \sum_{k=0}^{N-1} x_1[k]\, x_2^*[((-k))_N]$$



No se qué hacer con $X_2^*[((-k))_N]$

b) $x_1[n] = \frac{1}{N} \dfrac{\dfrac{e^{j\frac{2\pi}{N}n} - e^{-j\frac{7\pi}{N}n}}{2j}}{\dfrac{e^{j\frac{\pi}{N}n} - e^{-j\frac{2\pi}{N}n}}{2j}} = \dfrac{e^{j\frac{7\pi}{N}n} - e^{-j\frac{7\pi}{N}n}}{e^{j\frac{\pi}{N}n} - e^{-j\frac{\pi}{N}n}}$



Escaneado con CamS